Appendix F
Physics of Life Cycles: Pregnancy, Injuries, Diseases and Disorders, Aging, and Dying

This appendix overviews the physics of the body that is described throughout the text for special conditions. Each section begins by referring to relevant examples in the text chapters and continues by describing how to use the index to find these discussions.

F.1 Pregnancy

The physics of several features of pregnancy, birth, and newborn infants described in the text includes how the balance of the expectant mother is affected by changes in body mass (in Chap. 2) and the vertebra structure of women (Chap. 2), how the cervix softens as birth approaches (Chap. 4), how muscles grow in the fetus (Chap. 5), how the metabolic rate of the expectant mother increases due to increases in her own metabolic needs and the need to form the fetus, placenta, and so on (Chap. 6), the blood circulation needs of the expectant mother and fetus (Chap. 8), the breathing needs of the mother (Chap. 9), and how pregnancy is a positive feedback process in the body (Chap. 13).

Check the index listing for "Pregnancy" for all references to pregnancy, birth, and the newborn infant.

F.2 Injuries

The advantages of using canes and having hip replacements to recover from some injuries, and lower back pain and shoulder dislocations are discussed (in Chap. 2). Injuries that can result from collisions, including those that can occur during athletic events and in cars, are analyzed (Chap. 3). Damage to structural components of the body, such as due to bone breaking and tendon damage, and how to avoid such

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damage are addressed, along with the types of injuries that can occur during sporting events (Chap. 4).

Check the index listing for "Injuries" for all references to general physical injuries, and "Sports injuries" for those that commonly result from athletic events. For injuries to the eyes, those due to electric shocks, and the like, look under "Diseases and Disorders."

F.3 Diseases and Disorders

References to the physical aspects of several diseases and disorders are made throughout the text, including to concussions and other types of damage due to collisions of the head (in Chap. 3), heart and circulation diseases and disorders, such as hardening of the arteries, heart failure, arrythmias, transient ischemia attacks, and aneurysms (Chap. 8), breathing disorders (Chap. 9), modes of hearing loss due to medical disorders and middle ear pathologies (Chap. 10), vision-related conditions, such as the formation of cataracts, retinal detachment/disease, glaucoma, keratoconus, color blindness, and impaired of visual acuity (Chap. 11), and the effects of electrical shocks (Chap. 12).

Check the index listing for "Diseases and Disorders" for references to diseases, general disorders, injuries other than those to structural components that occur during common motions (to the eyes, the nervous system, and so on), the effects of bullets, and so on.

Disorders and severe or unusual conditions experienced by astronauts during and after space flight are addressed, including how their bodies change (Chap. 1), "moonwalking" and the effect of sustained acceleration (Chap. 3), and high sound volume during liftoff (Chap. 10).

We note that some diseases and disorders are thought by many to result largely from environmental factors in modern human life, such as type 2 diabetes, lower back pain (Chap. 2), hypertension and heart disease (Chap. 8), myopia (near-sightedness) (Chap. 11), and so on; these are often called (noninfectious) *mismatch diseases* [7].

F.4 Aging

Features of aging described in the text relate to the physics aspects of physiology. As you advance in age, you may need a cane for stability or to prevent pain while standing or walking (in Chap. 2). You walk and run slower, with shorter strides (Chap. 3). Bone density decreases (osteoporosis) and the fracture thresholds in bones decrease because of this and the accumulation of microcracks and strains; the materials properties of other structural components worsen as well (Chap. 4). Hip replacements may be necessary due to accumulated damage (Chap. 4 problems). The vertebra shape changes, leading to a decrease in height and forward bending of the spine and a hunched posture. Overall muscle mass decreases and strength decreases due to this

and neural factors, the percentage of fibers in muscles that are fast twitch decreases, and reactions become slower (Chap. 5). Peak peformance of even those highly trained in running and swimming decreases, as do basal metabolic rates (Chap. 6), which, along with a less active lifestyle, can lead to increased body weight (because increases of the mass of body fat more than compensates for the decrease in muscle mass and with this the body shape becomes more endomorphic) (Chap. 1). (The good news is that it then becomes easier for the body to float (Chap. 7).) However, those over 70 years of age often lose weight [17].

With advancing age, blood pressure typically increases and plaque accumulates in arteries, which can lead to heart disease, the maximium recommended heart rate decreases, and the cardiac output, stroke volume and heart rate decrease (Chap. 8). Oxygen consumption and the difference in arterial and venous oxygen decrease (Chap. 9). Hearing becomes worse as threshold hearing levels increase (presbycusis) and the voice ages (presbyphonia) with increases in pitch and shakiness, and so on (Chap. 10). The ability of the eye crystalline lens to accommodate to focus on both near and far objects decreases (presbyopia), and the likelihood of eye diseases and disorders increases, including eye crystalline lenses becoming cloudy as cataracts develop (Chap. 11). Reaction times increase and reflexes worsen, in part due to slower nerve reactions times, and our ability to balance worsens (Chap. 12). Finally, as the components of the circulatory system falter greatly, the body's feedback and control systems can accelerate the final stages of the dying process (Chap. 13).

Although most of these features associated with aging might not seem to be very attractive, on the positive side aging leads to a greater experience base and wisdom, and something else that the author is not able to recall at this moment.

Check the index listing for "Aging" for all references to aging.

More information on aging can be found in the general overview [8] and reviews of the theories of aging [16] and the biology and physiology of aging [1, 11], and in reviews focusing on the performance of masters athletes [13, 14], movement accuracy [2], bone loss (osteoporosis) [9], the extracellular matrix [12], muscle performance (which is called sarcopenia) [3], energy balance [17], swimming performance [4], the cardiovascular system [5], lung function [10, 15], and temperature regulation [6].

F.5 Dying

A few aspects of dying are noted in the text, including the likelihood of fatality during head trauma due to car crashes, boxing, and so on (Chap. 3), fatal heart conditions (Chap. 8), death from electrical shocks (Chap. 12), and a common mode of death in very old people that is promoted by positive feedback in the body (Chap. 13).

The text also covers areas related to The Rule of Threes, which is a very rough guide for survival before dying. You can survive 3 min without air (without breathing due to drowning or asphyxiation, Chap. 9) (and also 3 min in icy water, Fig. 6.22), 3 h without shelter (with extreme exposure, Chap. 6), 3 days without water (dehydration), and 3 weeks without food (starvation-but with proper water and shelter, Fig. 6.14). Extreme limits before death can be shorter or longer for a given person and conditions.

Check the index listing for "Dying" for all references to dying; you may also want look under "Diseases and Disorders."

References

1. I.B. Abrass, The biology and physiology of aging. W. J. Med. **153**, 641–645 (1990)
2. E.A. Christou, M. Shinohara, R.M. Enoka, Fluctuations in acceleration during voluntary contractions lead to greater impairment of movement accuracy in old adults. J. Appl. Physiol. **95**, 373–384 (2003)
3. T.J. Doherty, Invited review: aging and sarcopenia. J. Appl. Physiol. **95**, 1717–1727 (2003)
4. A.J. Donato, K. Tench, D.H. Glueck, D.R. Seals, I. Eskurza, H. Tanaka, Declines in physiological functional capacity with age: a longitudinal study in peak swimming performance. J. Appl. Physiol. **94**, 764–769 (2003)
5. A.U. Ferrari, A. Radaelli, M.Centola, Invited review: aging and the cardiovascular system. J. Appl. Physiol. **95**, 2591–2597 (2003)
6. W.L. Kenney, T.A. Munce, Invited review: aging and human temperature regulation. J. Appl. Physiol. **95**, 2598–2603 (2003)
7. D.E. Lieberman, *The Story of the Human Body: Evolution, Health, and Disease* (Pantheon, New York, 2013)
8. D.L. Longo, A.S. Fauci, D.L. Kasper, S.L. Hauser, J. Jameson, J. Loscalzo (eds.), *Harrison's Principles of Internal Medicine*, 18th edn. (McGraw-Hill, New York, 2011)
9. E. Seeman, Invited review, Pathogenesis of osteoporosis. J. Appl. Physiol. **95**, 2142–2151 (2003)
10. G. Sharma, J. Goodwin, Effect of aging on respiratory system physiology and immunology. Clin. Interventions Aging **1**, 253–260 (2006)
11. G.C. Sieck, Physiology of aging. J. Appl. Physiol. **95**, 1333–1334 (2003)
12. F.H. Silver, D.DeVore, L.M. Siperko, Invited review: role of mechanophysiology in aging of ECM: effects of changes in mechanochemical transduction. J. Appl. Physiol. **95**, 2134–2141 (2003)
13. H. Tanaka, D.R. Seals, Invited review: dynamic exercise performance in masters athletes: insight into the effects of primary human aging on physiological functional capacity. J. Appl. Physiol. **95**, 2152–2162 (2003)
14. H. Tanaka, D.R. Seals: Topical review: endurance exercise performance in Masters athletes: age-associated changes and underlying physiological mechanisms. J. Physiol. **586**, 55–63 (2008)
15. W.M. Wahba, Influence of aging on lung function-clinical significance of changes from age twenty. Anesth. Analg. **62**, 764–776 (1983)
16. B.T. Weinert, P.S. Timiras, Theories of aging. J. Appl. Physiol. **95**, 1706–1716 (2003)
17. M.-M.G. Wilson, J.E. Morley, Invited review: aging and energy balance J. Appl. Physiol. **95**, 1728–1736 (2003)